WEG Water Electric Generator

BRIEFING FOR SENSOR & IOT MAKERS

A power source that sits inert in storage and switches on when water arrives.

WEG is a patented galvanic cell that uses water as its electrolyte. Dry, it does nothing. When water reaches it, it starts producing electricity. For flood and leak sensing, the event you want to detect is the thing that powers the sensor.

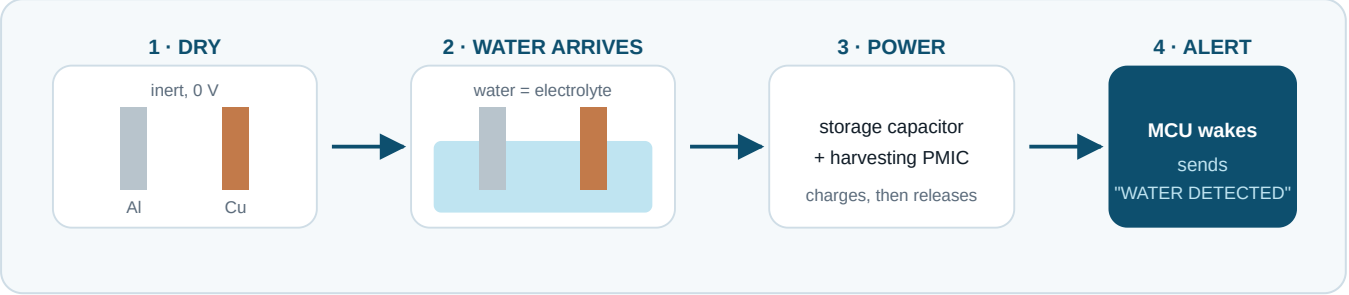
THE PROBLEM

- **Flood and leak sensors wait for years** for an event that may happen once. Their battery is drawn down the whole time, by standby current and self-discharge.
- **Battery checks and replacements** are a recurring field cost across thousands of installed units, and they get skipped.
- **A sensor that is flat on the day of the flood** is a failed product, an insurance claim and a lost customer.
- **Lithium primary cells** add cost, shipping rules and end-of-life waste to low-cost devices.

HOW WEG WORKS

Each cell pairs an **aluminium anode** with a **copper cathode**. Water, plain or salty, closes the circuit as the electrolyte, and the aluminium is slowly consumed to produce current.

- **Dry storage:** no reaction, so no self-discharge before deployment.
- **Wet:** output starts within seconds and continues while water is present.
- **Materials:** aluminium and copper. No lithium, no acids, no heavy metals.
- **Scales by cells:** cells in series raise voltage; more area raises current.



~3 V
from six 10 × 2 cm cells in plain water, as measured in the patent

µA class
continuous current: a microwatt source, designed for long life, not high power

Inert dry
no electrolyte in the cell until water arrives, so no reaction in storage

Al / Cu
common, low-cost, non-toxic materials; plain or salt water

WHERE IT FITS

Application	Why a water-activated source helps
Flood beacons and river sensors	Water arrival is both the trigger and the power source. No battery to check between floods.
Building leak detectors	Basements, risers and under-sink units that sit dry for years and must work on the one day it matters.
Insurance-linked sensors	A sensor whose reliability does not depend on a battery replacement schedule.
Wet-environment trickle charging	Topping up a capacitor or small cell in permanently wet sites (soil, tanks, marine), where light and heat harvesting are not available.

The demo: water in, power out

In our demo video, a rolled WEG cell sits dry in a clear bowl. Water is poured in, and an LED wired to the cell lights up within about two seconds and stays lit. That power-on moment is exactly what a flood or leak alarm needs: the arrival of water is the signal.

1. **Dry**
   The cell sits in an empty bowl. The LED is off.
2. **Water in**
   Plain water is poured over the cell.
3. **Power**
   The LED lights within about two seconds and stays on.
4. **Next: alert**
   ESP32-S3 firmware sends "WATER DETECTED" on power-up.

Watch the demo: youtu.be/lEEah8G4E_8
The power-on-equals-alarm logic needs no sensing circuit at all: any MCU that boots can report the event. In a product design, a harvesting PMIC and a small capacitor buffer the radio's transmit peaks.

PATENT PROTECTION

EP 3 039 172

Type	European patent, granted 2019
In force in	Germany, United Kingdom, France, Italy, Sweden (fees paid 2025)
Term	to approximately 2034
Ownership	Co-owned by the inventors; licensing and joint development are open for discussion

WHAT WE CAN SAY TODAY

- **Measured:** about 3 V from six cells in plain water, at microamp-level current, as recorded in the patent.
- **Demonstrated:** an LED lit by the cell within seconds of water arriving (see the video). ESP32-S3 alarm firmware is ready for testing in pilots.
- **Under way:** independent measurements with a university partner, covering output over time in different waters.
- **Not claimed:** high power or unlimited life. The aluminium anode is consumed while wet; life depends on cell size and load.

PILOT OFFER FOR SENSOR COMPANIES

A few partners test WEG in their own hardware, judged by their own engineers.

1. **Intro call (20 min):** your device, its power budget and where water is present.
2. **Mutual NDA,** then cell construction details and evaluation samples.
3. **60 to 90 day pilot** in your device, with success criteria we agree up front (for example time to first alert, output over 30 days, behaviour in tap, rain and salt water).
4. **Next step on success:** a joint reference design, a development agreement or a licence under EP 3 039 172.

Botond Bereczki · Business Development, WEG
weg.sensor@gmail.com · [phone]

Figures above are from the patent's own measurements and our demonstrator. Product-level performance will be confirmed in joint testing.